Module 3

Introduction to Statistical Analysis and EHR Data

Cynthia Sandor

Edmond J Safra Assistant Professor
UKRI Future Leader Fellow
UK Dementia Research Institute Group Leader

UK Dementia Research Institute
Imperial College London
THE MICHAEL J. FOX FOUNDATION FOR PARKINSON'S RESEARCH
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What We'll Do Today

- **Lecture:** Short introduction to EHR data in Parkinson's: UPDRS III and LEDD
- **Lecture:** Introduction to Statistics
- **Tutorial 3 / Challenge, with Jared Maina**: prepare real clinical data for analysis, then test it

Part 1: visualise distributions and relationships, flag outliers and data-entry errors, resolve duplicates and handle missing data.

Part 2: choose and run statistical tests, check their assumptions, compare groups, measure associations, correct for multiple comparisons and report the results.

Lecture Plan

1 EHR in Parkinson's

- Electronic health records are tables
- MDS-UPDRS Part III and LEDD
- One row per clinic visit

2 Describe and clean

- Categorical and continuous data
- Centre, spread and shape
- Skew and log scales
- Outliers and data-entry errors
- Duplicates
- Missing data

3 Test and report

- What a test does: the p-value
- Parametric or non-parametric?
- Comparing groups
- Associations
- Multiple comparisons
- Effect sizes, confidence intervals, reporting

What an EHR table looks like

patient_id	sex	age	ethnicity	BMI	years_since_dx	updrs_iii	LEDD
P0005	Female	71.2	White	29.9	1.60	27.9	936.7
P0005	Female	72.7	White	29.9	3.07	30.3	639.5
P0012	Female	55.5	White	21.5	6.89	**NaN**	**NaN**
P0030	Female	62.8	**(blank)**	30.3	2.64	22.7	1275.5
P0586	Female	**178.0**	White	22.9	7.68	**NaN**	2221.2
P0148	Male	49.3	White	**201.0**	0.69	15.2	1421.1
P0421	Male	76.0	White	25.0	3.52	**187.6**	1158.7

Blue: the same patient at two visits. **Red: missing, blank or impossible values.**

Electronic health records are tables: one row per clinic visit, one column per variable. The same patient appears several times, and real records contain gaps and errors.

The Parkinson’s measures in the data

MDS-UPDRS Part III

- A clinician-rated assessment of motor symptoms.
- Scores run from 0 to 132: higher means more severe.
- Severity bands: mild 0–32, moderate 33–58, severe 59 or more.
- A score outside 0–132 is an error.

LEDD (mg/day)

- Levodopa equivalent daily dose: the total daily dose of Parkinson’s medication.
- Different drugs are converted to a common scale, so they can be compared.
- 0 means no medication.
- Skewed: doses escalate as Parkinson’s progresses.

LEDD trajectory calculation methodology

1) Calculate LEDD for each prescription
   1) $LEDD = daily_dose * substance_strength * conversion_factor$
   2) e.g. LEDD (controlled-release levodopa):
      4 tablets a day * 125 mg * 0.75 = 375
   3) Some medication have fixed LEDD despite of the dose
      => LEDD(Safinamide) = 150
2) Establish the end date for each prescription
3) For each time point, sum the LEDDs for active prescriptions at this timepoint
   1) COMT do not contribute to summation
4) For each timepoint, If COMT prescription(s) are active at this time point, multiply by product of COMT conversion factors

Table 1. Conversion factors used to calculate LEDDs for PD drugs

Drug Class	Drug	Conversion factor
Dopamine Replacement	Levodopa	DD x 1
	Dual-release levodopa (Madopar DR)	DD x 0.85
	Controlled-release levodopa	DD x 0.75
	Extended-release levodopa (Rytary)	DD x 0.5
	Intrajejunal levodopa/carbidopa infusion	DD x 1.11
	Intrajejunal levodopa/carbidopa/entacapone infusion	DD x 1.11 (morning) + DD x 1.46 (maintenance and extra doses)
	Subcutaneous foslevodopa/foscarbidopa	DD x 0.75
	Inhaled levodopa (Inbrija)	DD x 0.69 (capsules)
COMT Inhibitors*	Entacapone	LD x 0.33
	Tolcapone	LD x 0.5
	Opicapone	LD x 0.5
Dopamine Agonists	Pramipexole, Lisuride, Pergolide	DD x 100
	Ropinirole	DD x 20
	Rotigotine	DD x 30.3
	Piribedil	DD x 1
	Apomorphine injected	DD x 10
	Apomorphine sublingual (Kynmobi)	DD x 1.5
	Bromocriptine	DD x 10
	Cabergoline	DD x 66.7
	Dihydroergocryptine (DHEC)	DD x 5
MAOB Inhibitors	Selegiline oral	DD x 10
	Selegiline sublingual	DD x 80
	Rasagiline	DD x 100
Other	Amantadine	DD x 1
	Amantadine ER (Gocovri)	DD x 1.25
	Amantadine ER (Osmolex)	DD x 1
	Safinamide (Xadago)	LED = 150
	Zonisamide	LED = 100
	Trihexyphenidyl	LED = 100
	Istradefylline (Nourianz)**	LD x 0.2
	Mucuna pruriens	DD x 0.013

DD = daily dose of drug being converted; LD = levodopa dose; LED = levodopa equivalent dose.

Why Statistics?

- **Real clinical data are messy:** describe and clean them before you analyse them.
- **Then test:** could a difference this large easily arise by chance?

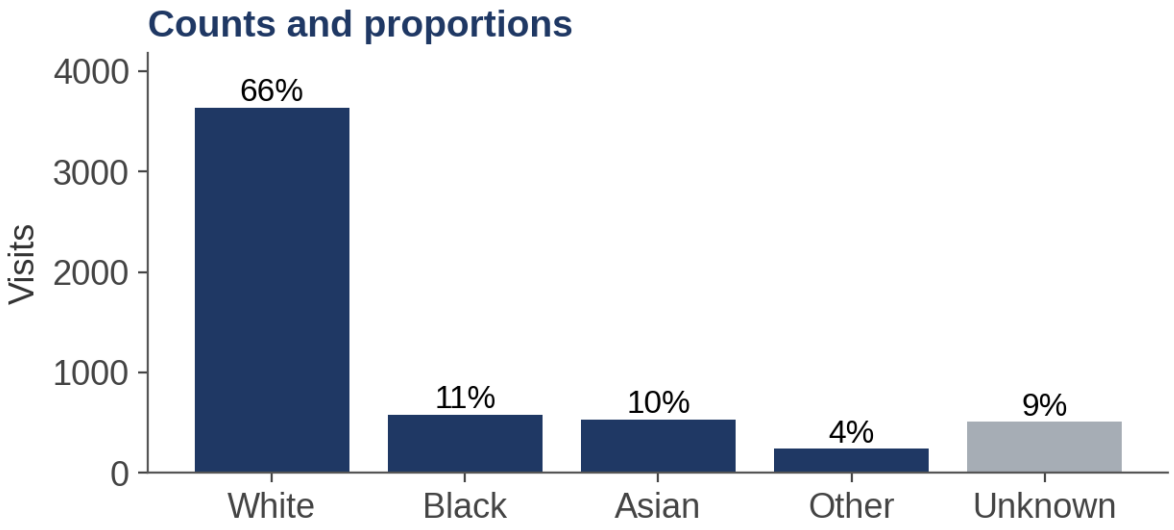
Key questions:

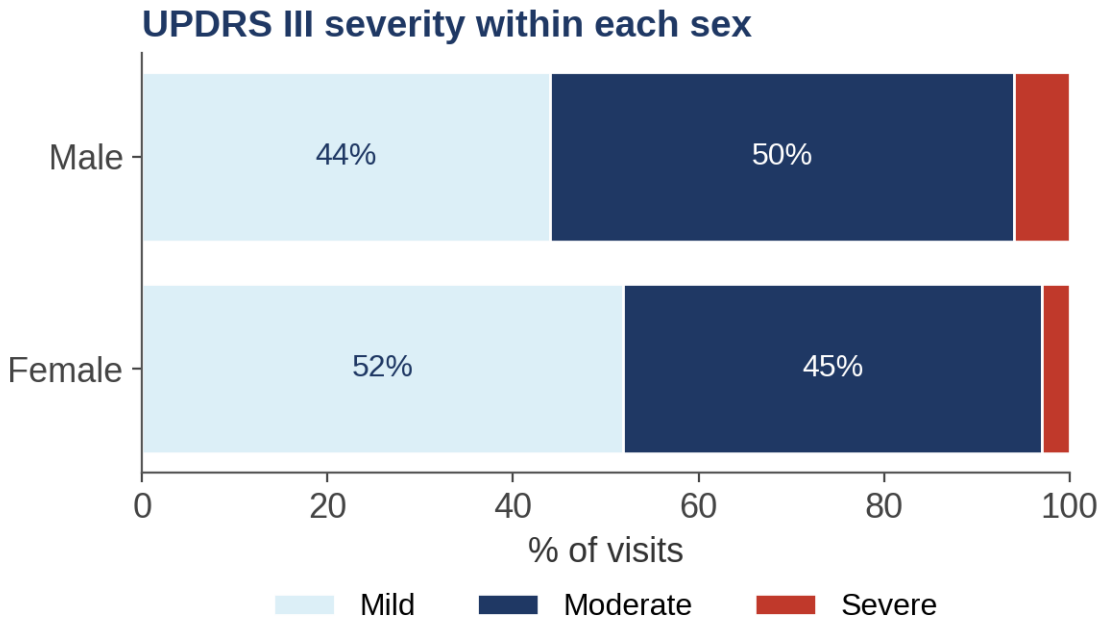
- Do groups differ?
- Are two variables related?
- Is a difference real, or just chance?

What are Descriptive Statistics?

- Summarize and describe data
- First step before testing hypotheses
- Measures of central tendency & variability

Categorical data: counts and proportions

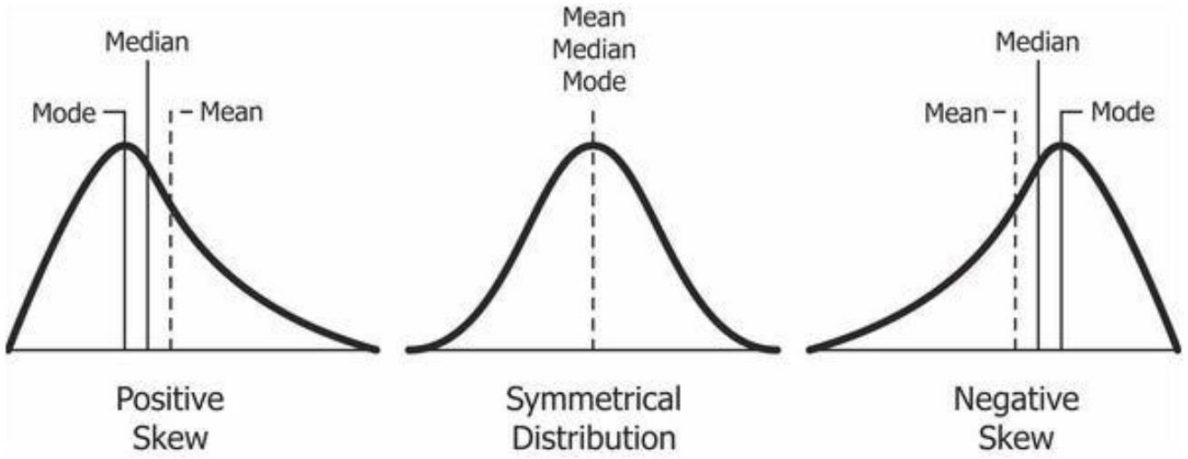




Count every category, including an explicit “Unknown”: hiding missing labels distorts the percentages.

Measures of Central Tendency

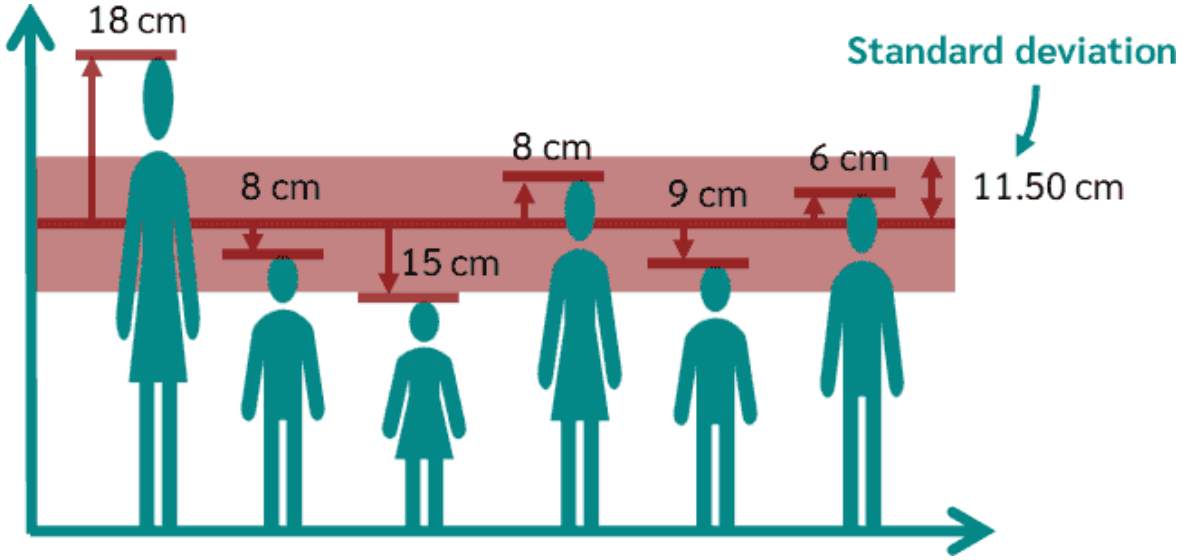
- **Mean:** Sum of all values divided by the total number of values
- **Median:** Middle value (when the data is arranged in order)
- **Mode:** Most common value (the value with the most frequency)



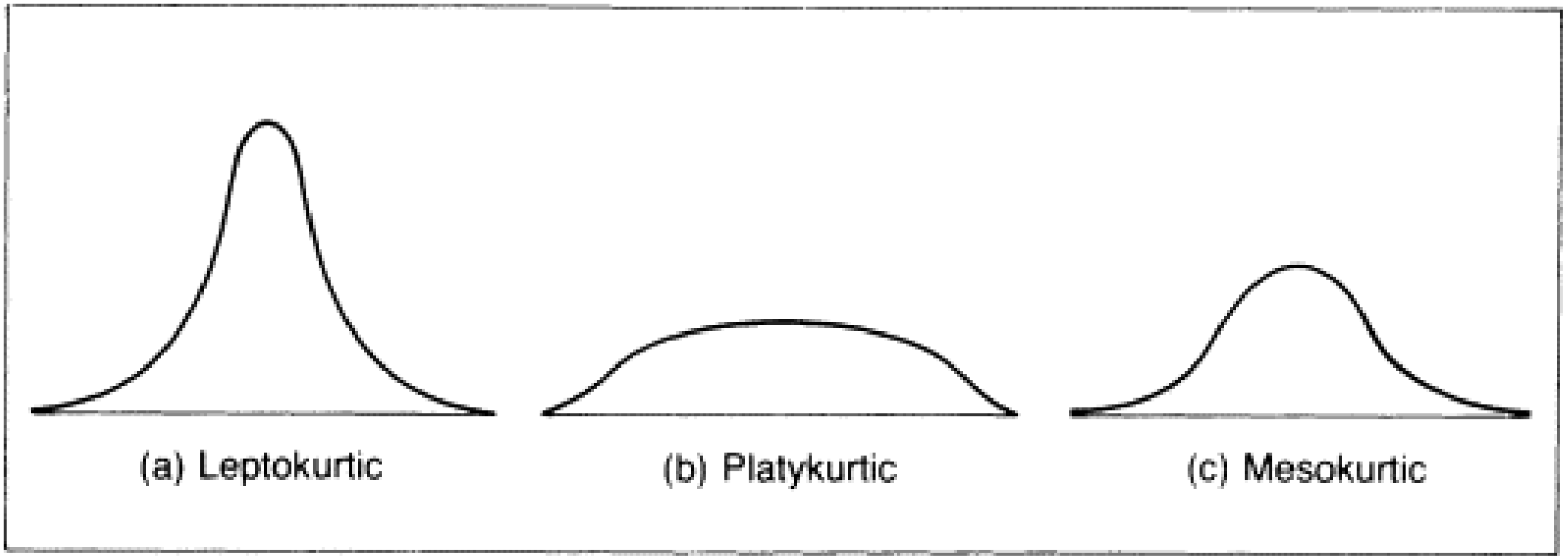
Variability

- **Range** = Highest value − Lowest value
→ Quick measure of spread
- **Variance (Var)** = Average squared distance from the mean

- **Standard Deviation (SD or σ)** = Square root of variance
→ “Average distance” of values from the mean

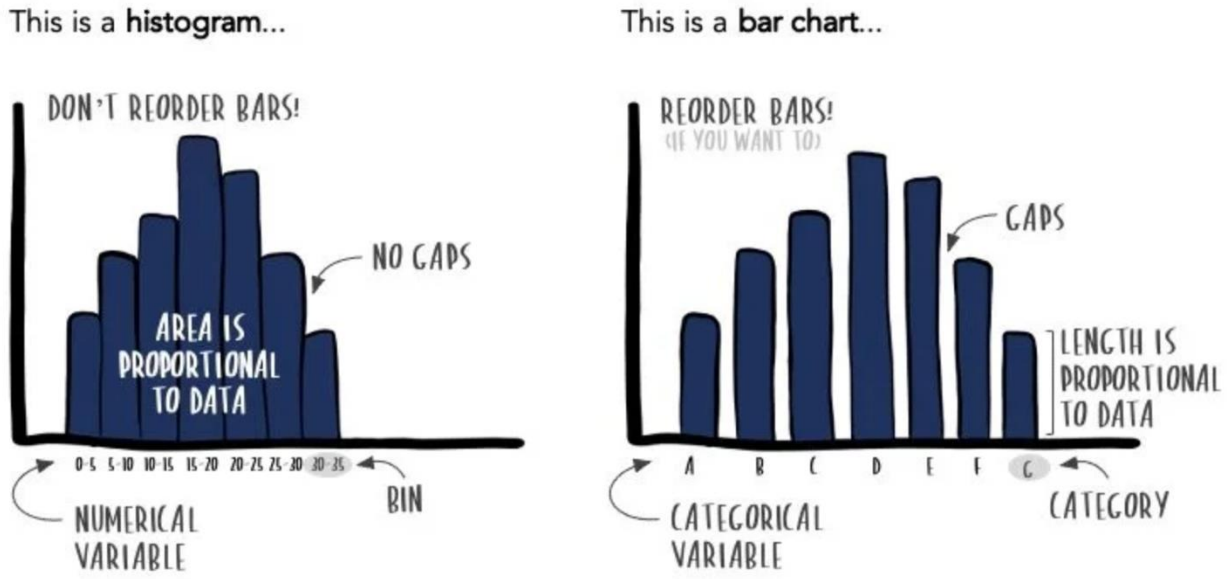


Kurtosis



Visualisation Data

- **Histogram** → shows how the values of a continuous variable are spread
- **Bar chart** → compares counts of categories
- **Box and violin plots** → compare distributions between groups
- **Scatter plot** → shows the relationship between two continuous variables



Visualisation Data

Boxplot (five-number summary):

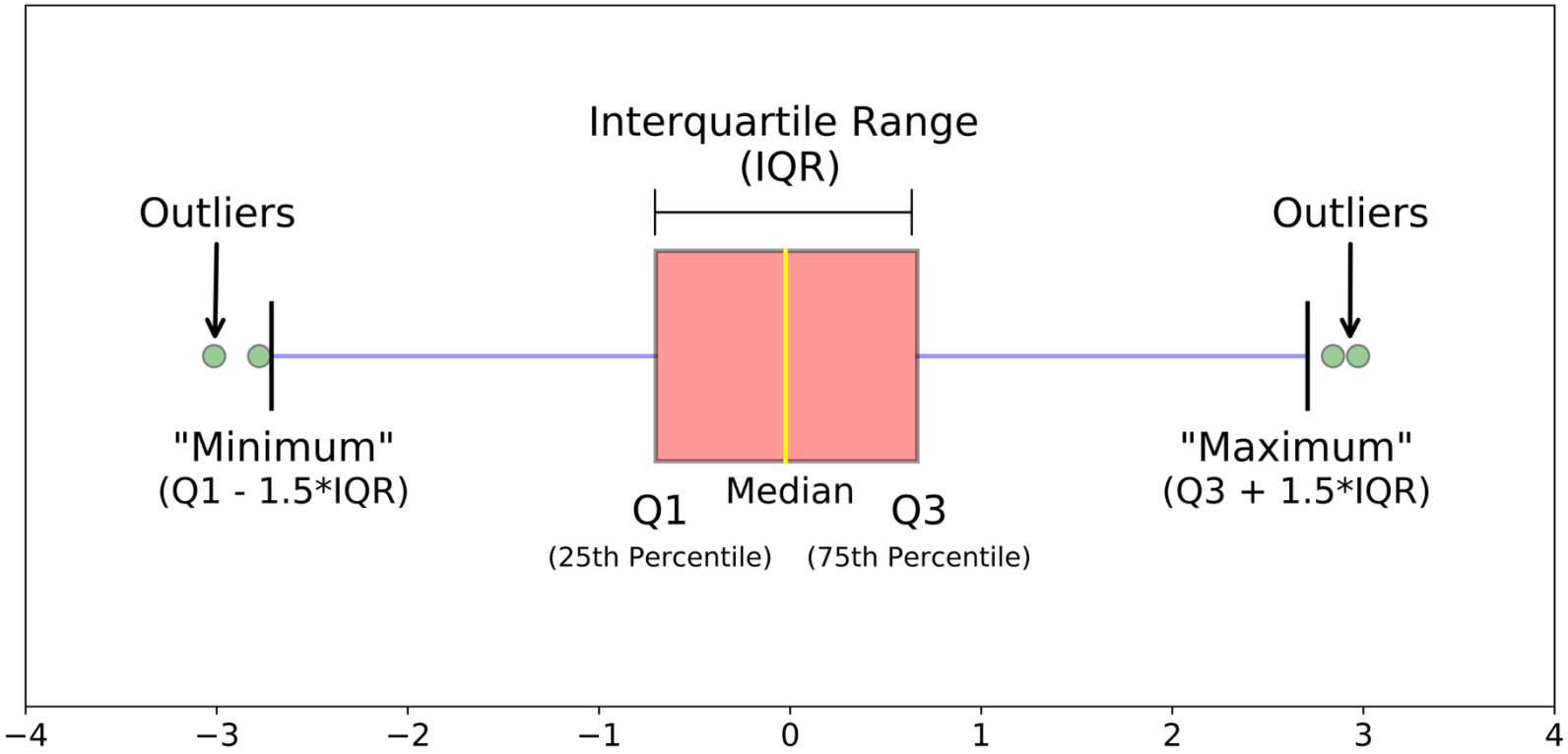
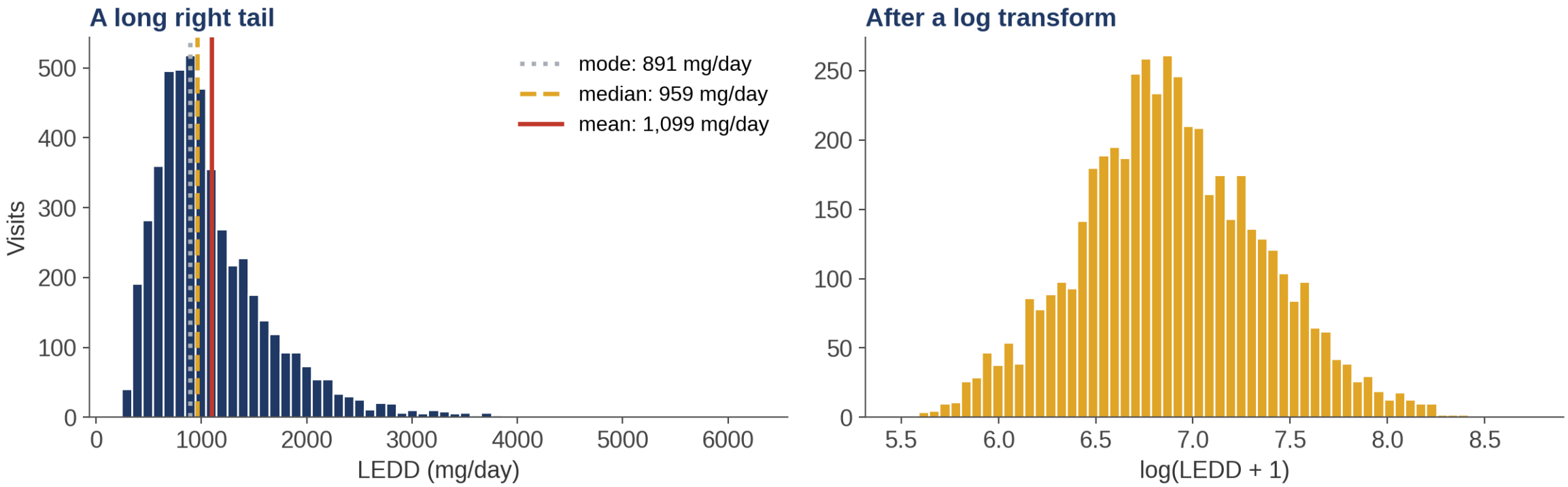


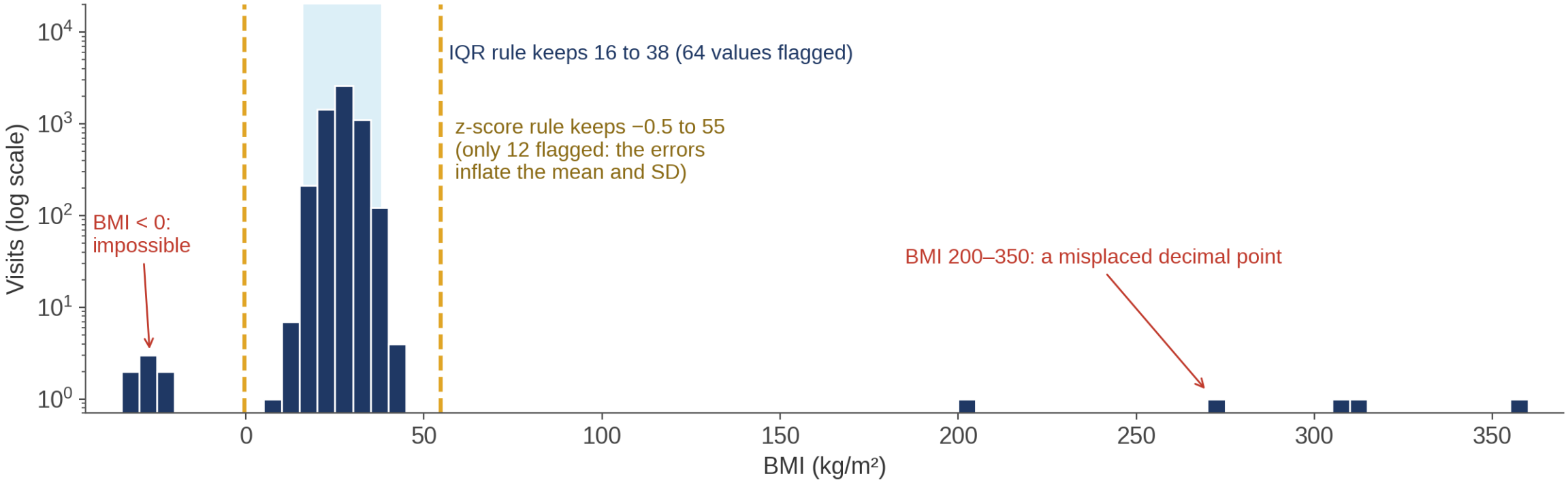
Table of summary statistics: Mean, Median, SD, etc.

Skewed data and log scales



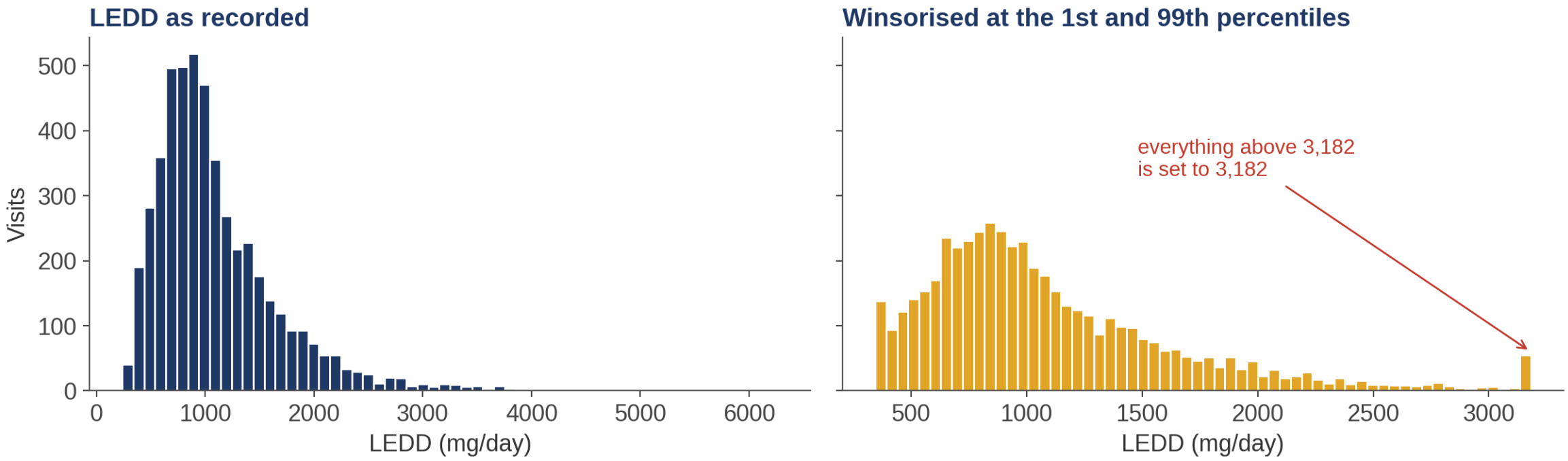
In a right-skewed variable such as LEDD, the mean is pulled above the median; a log transform makes the shape roughly symmetric.

Outliers: error or genuine extreme?



Rules only flag unusual values. Deciding whether each one is an error or a real patient takes clinical judgement: flag, decide, document.

Capping, winsorising or setting to missing



Capping limits values at a fixed maximum, such as the top of a scale. **Winsorising** caps at percentiles.

An impossible value is an error, not an extreme: set it to missing rather than inventing a score.

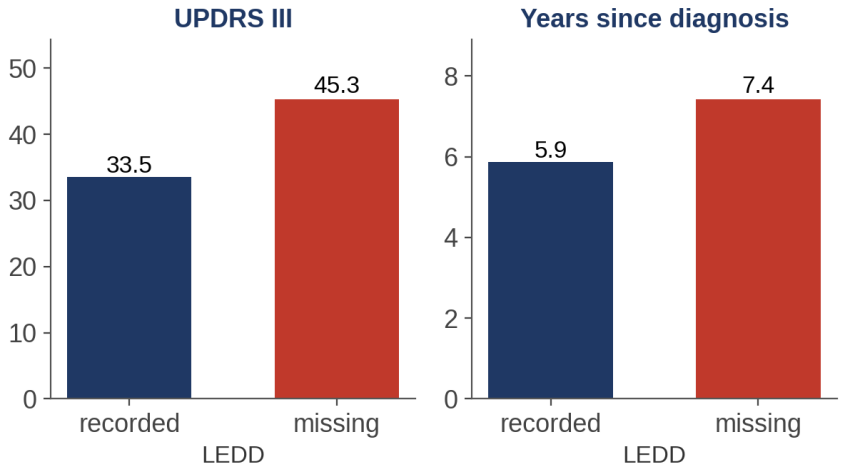
Duplicates: three different things

Type	What it looks like	What to do
Exact duplicate	Every column matches another row: a copy-paste or merge artefact	Drop it
Key duplicate	Same patient and same visit time, but slightly different values	Inspect it, then apply a documented rule (keep the first, or take the median)
Repeat visit	Same patient at a different time	Keep it: these are your repeated measures

In medication records, two identical rows on the same day may be a genuine prescription: always interpret duplicates in the context of the data source.

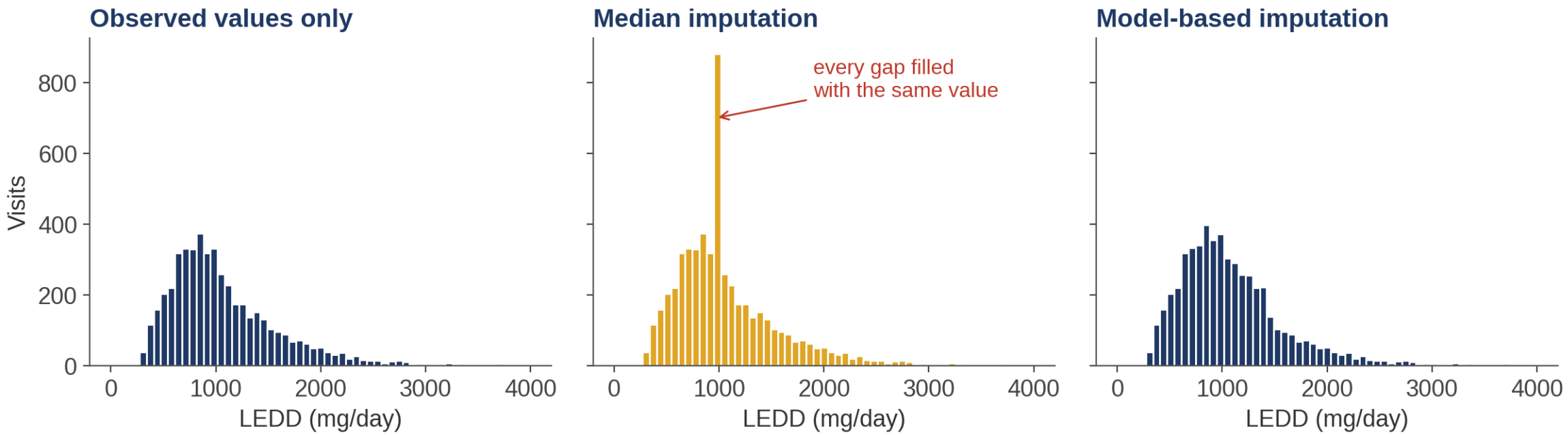
Missing data: why is it missing?

	Missingness depends on	Example	Consequence
MCAR Missing completely at random	nothing	a lab analyser fails at random	complete cases are unbiased, just fewer
MAR Missing at random	other observed variables	UPDRS III skipped more often at one clinic site	correct by imputing from those variables
MNAR Missing not at random	the missing value itself	the most severe patients stop attending	cannot be fixed from the data alone



Visits with missing LEDD have higher UPDRS III scores and longer disease duration: LEDD is not missing completely at random.

Handling missing data



Drop rows only when little is missing at random. Median imputation creates an artificial spike; model-based imputation keeps the relationships between variables.

Quick Check: Describe and Clean

Q1

Reaction times from 5 students: 220, 230, 400, 240 and 250 ms.

What is the mean? The median? Which is more robust to the outlier (400 ms)?

Q2

A patient's BMI is recorded as 201.

Error or genuine extreme? What do you do with it?

Q3

LEDD is missing more often for patients with higher UPDRS III scores.

Is it missing completely at random (MCAR), at random (MAR) or not at random (MNAR)?

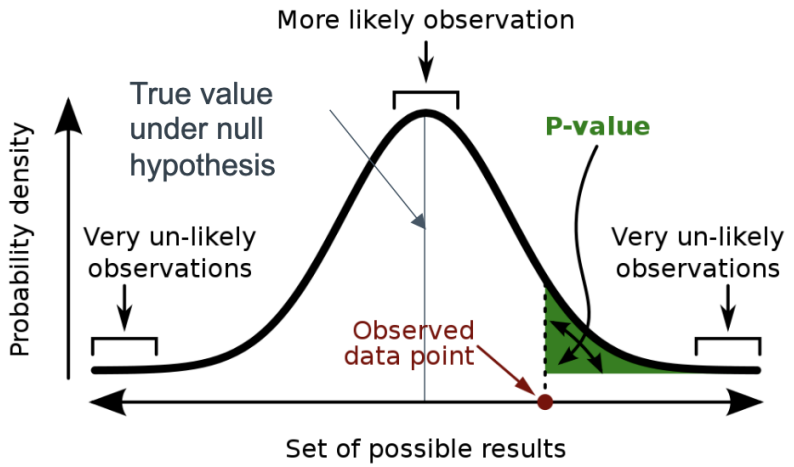
Hypothesis Testing & P-values

- **A hypothesis test asks:** could a difference this large easily arise by chance?
- **Null hypothesis (H_0):** there is no real difference.
- **Null distribution:** the differences chance alone usually produces.
- **p-value:** the proportion of those chance results at least as extreme as what we observed.
- **If $p < 0.05$ (α, a convention, not a law):** statistically significant, so we reject H_0.

Hypothesis Testing & P-values

What is the probability (P) of observing the sampled data under the null hypothesis? • If P is less than a selected threshold α (often 0.05), the null hypothesis is rejected – Statistical significant

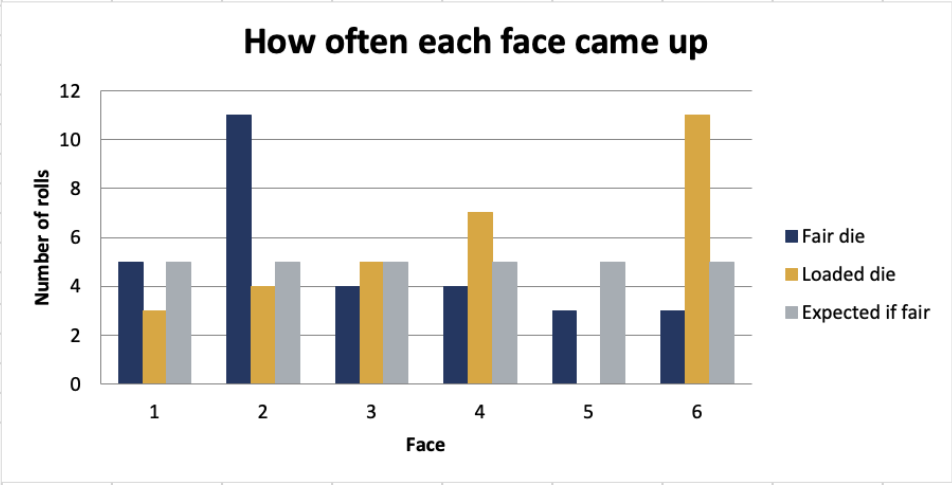
The P value is defined as the probability, under the null hypothesis H0, of obtaining a result equal to or more extreme than what was actually observed – P(result | H0).



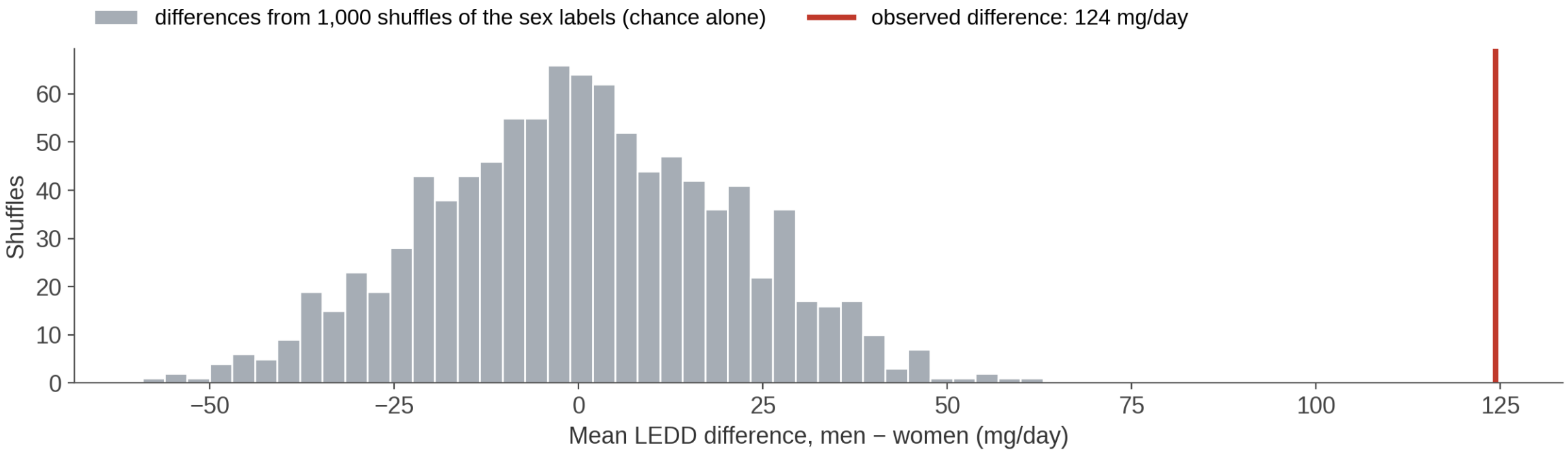
A **p-value** (shaded green area) is the probability of an observed (or more extreme) result assuming that the null hypothesis is true.

Hypothesis Testing & P-values

	A	B	C	D
1	**Is this die loaded towards 6?**			
2	*Yellow cells are inputs: change them, then press F9 to roll again.*			
3				
4	**Number of rolls (1 to 50)**	**30**		
5	**Chance of a 6 on the loaded die**	**50%**		
6	**Chance of a 6 on a fair die**	16.7%		
7	**Significance level, α**	**0.05**		
8				
9	H0: the die is fair, so a 6 comes up 1 time in 6.			
10	p-value: if the die were fair, the chance of at least as many sixes as we saw.			
11				
12	**Result**	**Fair die**	**Loaded die**	
13	**Number of rolls**	30	30	
14	**Sixes observed**	3	11	
15	**Sixes expected if the die were fair**	5.0	5.0	
16	**p-value**	0.8972	0.0067	
17	**Decision at α**	Do not reject H0: no evidence the die is loaded	Reject H0: evidence the die is loaded	
18				
19				
20	**Face**	**Fair die**	**Loaded die**	**Expected if fair**
21	**1**	5	3	5.0
22	**2**	11	4	5.0
23	**3**	4	5	5.0
24	**4**	4	7	5.0
25	**5**	3	0	5.0
26	**6**	3	11	5.0
27				
28				
29				
30				



A p-value, by shuffling



Shuffle the labels many times to see the differences chance alone produces. The p-value is the share of shuffles at least as extreme as the real difference.

Errors in Hypothesis Testing

- **Type I error (false positive):** reject H_0 when it's true

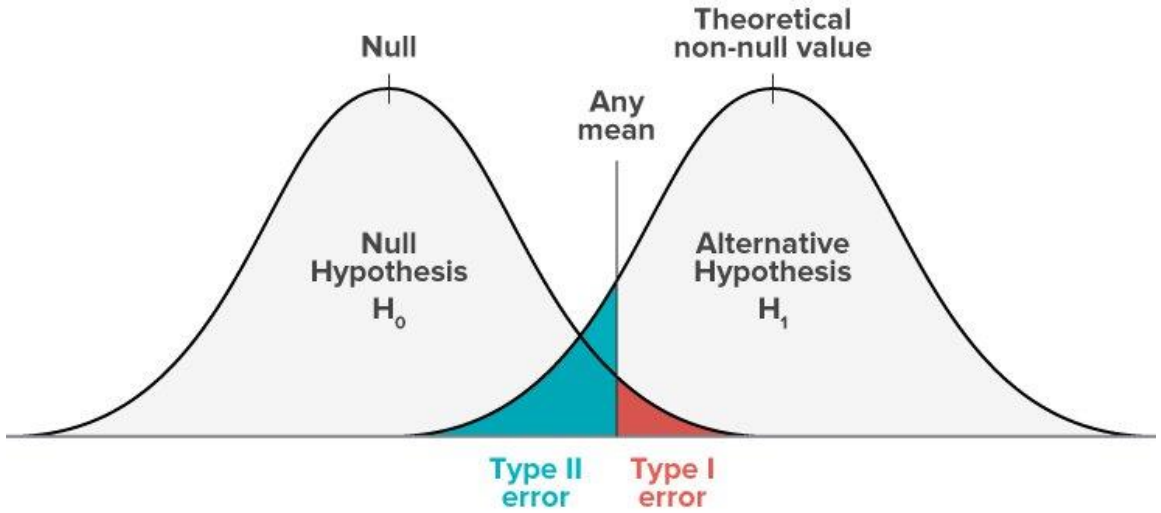
Probability = α (significance level, usually 0.05)

- **Type II error (false negative):** fail to reject H_0 when H_1 is true

Probability = β

- **Power = 1 − β** = chance of correctly detecting a true effect

Balancing errors is key to good study design



Parametric vs. Non-parametric Tests

- **Parametric tests**

  Compare means and assume a roughly normal shape

  Welch’s t-test, one-way ANOVA, Pearson correlation

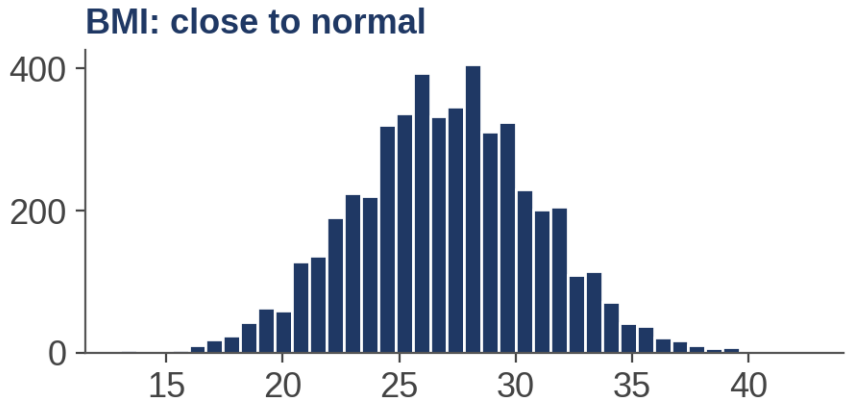
- **Non-parametric tests**

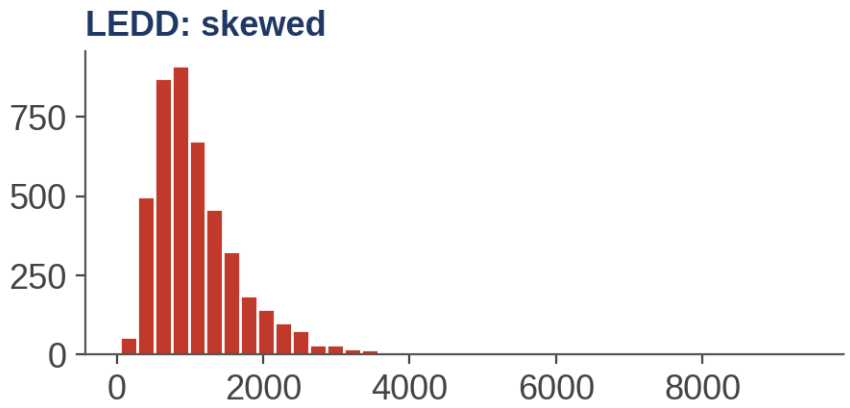
  Compare ranks and make no shape assumption

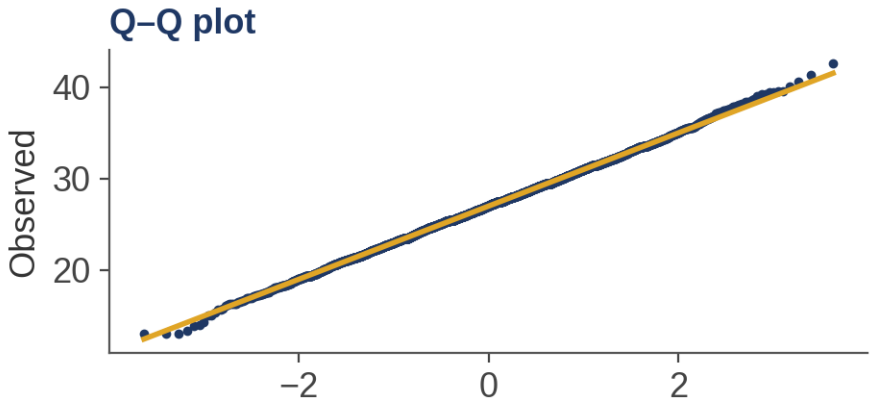
  Mann–Whitney U, Kruskal–Wallis, Spearman correlation

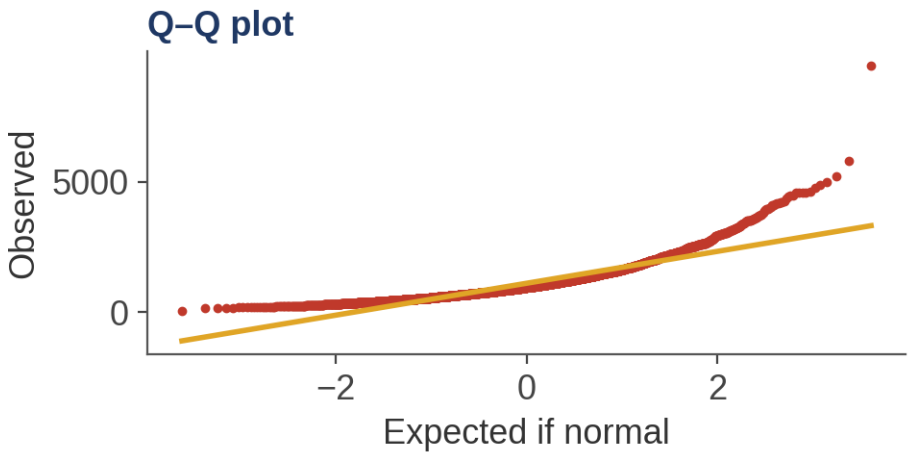
- **Look first, test second:** if the data are skewed or full of outliers, the mean is a poor summary, so use the non-parametric test.

Checking assumptions: is the data normal?









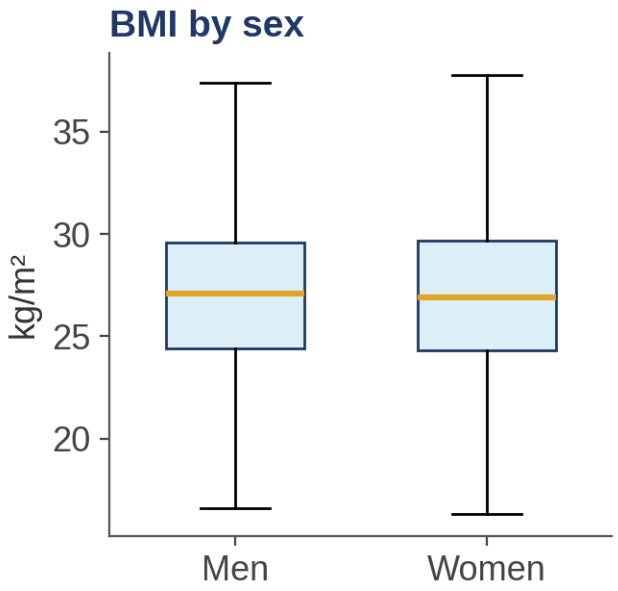
- Points on the line: normal. BMI follows it; LEDD bends away.
- In large samples, formal tests (Shapiro–Wilk, Levene) flag trivial deviations: trust the plots.
- Tests assume independent observations: one row per patient.

Choosing the right test

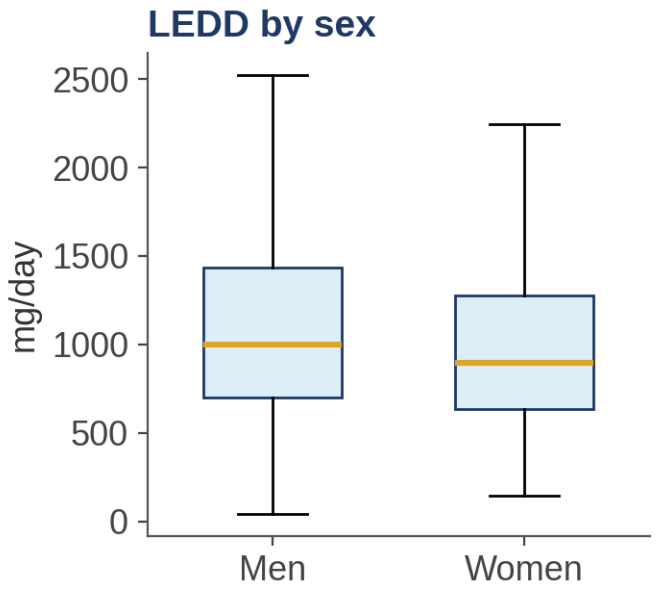
Question	Parametric	Non-parametric
Do 2 groups differ?	Welch’s t-test	Mann–Whitney U
Do 3 or more groups differ?	One-way ANOVA	Kruskal–Wallis
Are 2 numeric variables related?	Pearson correlation	Spearman correlation
Are 2 categorical variables related?	Chi-squared test	Fisher’s exact test (small counts)

The same patients measured twice, before and after treatment? The observations are not independent: use the paired versions.

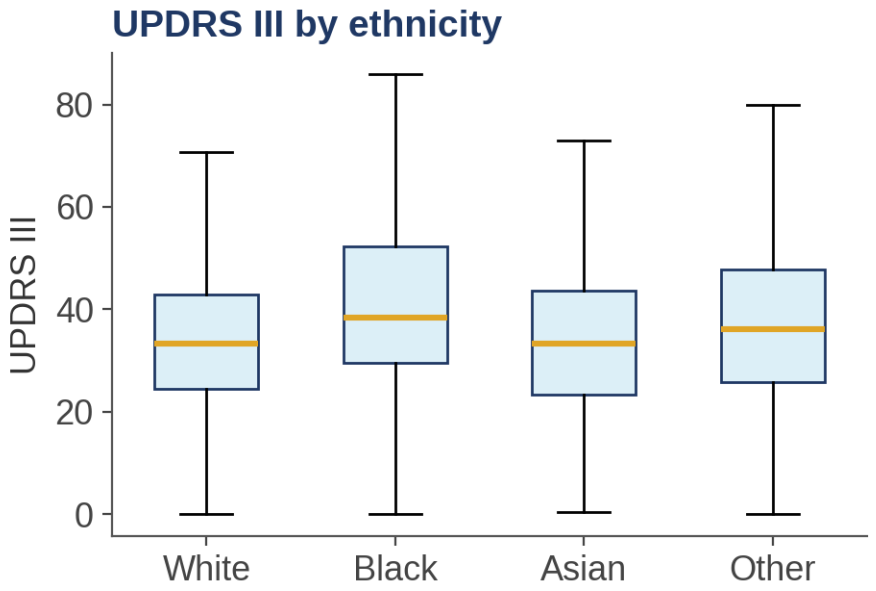
Comparing groups: means or ranks?



Welch's t-test: p = 0.48
difference 0.08 (95% CI −0.14 to 0.31)



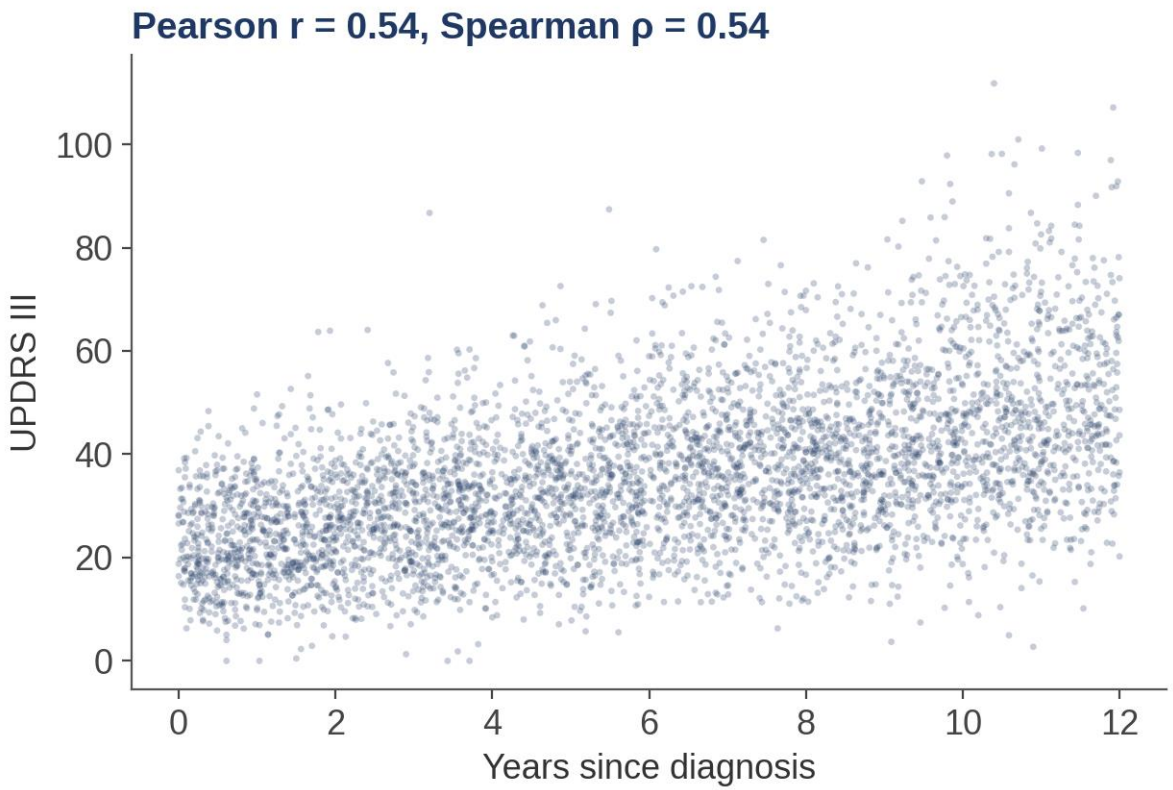
Mann–Whitney U: p < 0.001
medians 997 vs 895 mg/day



one test for all groups: ANOVA or Kruskal–Wallis, p < 0.001
(at least one group differs, not which one)

Normal data: compare means (Welch's t-test, ANOVA). Skewed data: compare ranks (Mann–Whitney U, Kruskal–Wallis).

Associations: correlation and chi-squared



UPDRS III severe (59 or more), by sex

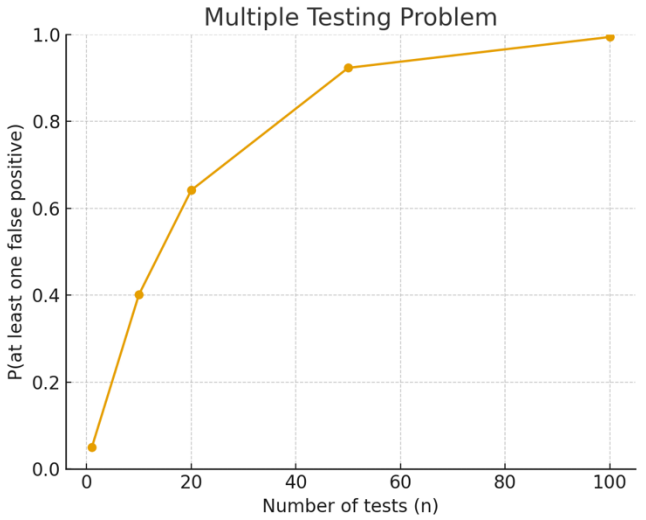
	Severe	expected	Not severe	expected
Male	247	192	2181	2236
Female	105	160	1914	1859

Men have more severe scores than expected if sex and severity were unrelated: χ^2 test, $p < 0.001$.

Correlation shows how two numbers move together, not what causes what. Chi-squared compares observed counts with those expected if two categories were unrelated.

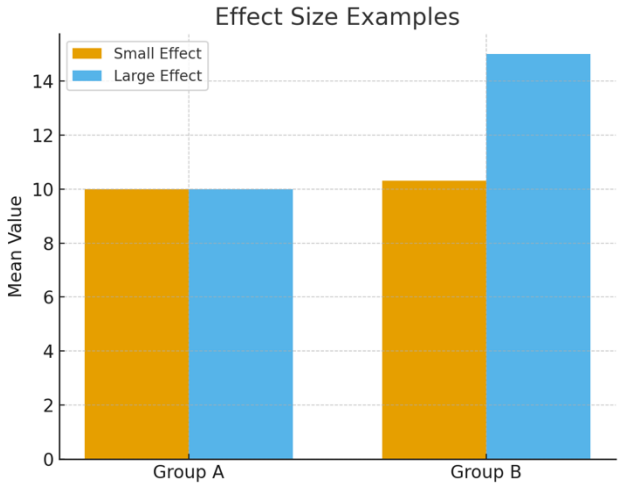
Multiple Testing Problem

- Every test at α = 0.05 carries a **5% risk of a false positive**
- Run 20 tests where nothing is going on: **expect one "significant" result by chance**
- A correlation matrix of 10 variables already contains 45 tests
- **Bonferroni:** controls the chance of any false positive; strict, for a few pre-planned tests
- **Benjamini–Hochberg:** controls the proportion of false positives; for many tests
- Choose the correction before looking at the results, and report it



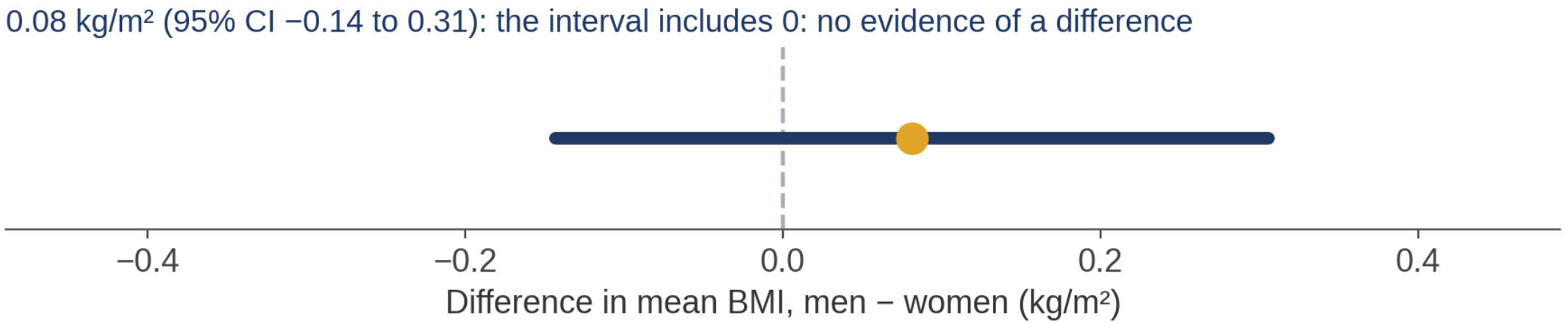
Effect Size

- A p-value says whether a difference could be chance, **not how big it is**
- A small p does not mean the effect is large or clinically important
- With thousands of patients, tiny differences become “significant”
- Always report the effect size in the data’s own units, **such as the difference in means in mg/day, with its 95% confidence interval**



Confidence Interval

A 95% confidence interval is the range of plausible values for the true effect. **A 95% CI that excludes 0 matches $p < 0.05$.**



Reporting your results

Mean BMI did not differ between men and women (27.0 vs 26.9 kg/m²; difference 0.08, 95% CI −0.14 to 0.31; Welch's t-test, p = 0.48; n = 2,717 and 2,222).

Every result should say:

- which test you used, and why
- how many patients were in each group
- the centre and spread: mean and SD, or median and IQR
- the effect size with its 95% confidence interval
- the exact p-value (or p < 0.001)
- any correction for multiple comparisons

Quick Check: Test and Report

Q1

You compare LEDD between men and women and get p = 0.01.

What is the null hypothesis? What does p = 0.01 mean?

Q2

LEDD is strongly right-skewed.

Which test would you use to compare men and women?

Q3

You run 20 tests where nothing is really going on.

How many “significant” results do you expect by chance?

Take-home Messages

- **Visualise your data first**: plots reveal errors and shapes that tables hide
- **Clean before you test**: flag, decide and document outliers, duplicates and missing data
- **Look first, test second**: choose a parametric or non-parametric test from the shape of the data
- **A p-value is not the whole story**: report effect sizes, 95% CIs and any correction for multiple comparisons

Let’s Code



Tutorial 3: two notebooks, in this order

1. pandas_ehr_tutorial_part1_2026_exercises.ipynb: visualise and clean
2. pandas_ehr_tutorial_part2_2026_exercises.ipynb: statistical analysis

https://github.com/Sandoretal/Module_3/tree/main/tutorial_3